SWE 363 · Assignment 4

Software Engineering Portfolio

From Local to Live

Aseel Bawazir
Student ID: 202270080
King Fahd University of Petroleum & Minerals (KFUPM)

Live: seel88/202270080-AseelBawazir-assignment4

Project Vision & Goals

Objective To design and deploy a production-ready web application that showcases academic proficiency and engineering domain expertise.

🎨 Professional Branding

Career-ready portfolio design reflecting industry standards.

⚡ Async Logic Mastery

External API consumption via GitHub REST API integration.

♿ Accessibility & Persistence

High-contrast UX with localStorage theme persistence.

🚀 CI/CD Live Deployment

Transition from static code to automated GitHub Pages builds.

The Technical Stack

🌐 **Frontend Core**	HTML5 (Semantic) · CSS3 (Grid/Flexbox)	MARKUP
⚙️ **Logic Engine**	Vanilla JavaScript ES6+	SCRIPT
🔗 **Data Source**	GitHub REST API · Live repo sync	API
💾 **State Management**	Web Storage API · localStorage	STORAGE
🚀 **Deployment Pipeline**	GitHub Pages · GitHub Actions CI/CD	DEVOPS

Engineering Implementation

🔗 GitHub API Integration

```
fetch(
  'api.github.com/users/seel88/repos')
```

Dynamic Fetching
Live repo data pulled from GitHub REST API on every visit. No static data.

Custom Filtering
Intelligent filtering to display only relevant personal and academic projects.

💾 Persistent State Management

```
localStorage.setItem(
  'theme', currentTheme)
```

Session Memory
App remembers user's dark/light theme preference across browser sessions.

Zero-Lag Application
Theme loads via immediate script execution — no flash of unstyled content.

Innovation Feature: The Typewriter

Feature: **Custom JavaScript Typewriter Animation**

portfolio.js

```javascript
function typeWriter() {
  const roles = [
    "SE Student",
    "Full-Stack Dev",
    "Testing Eng"
  ];
  setTimeout(
    deleteChar, delay
  );
}
```

🔁 Recursive Architecture

Self-referencing function loop creates seamless role rotation.

⏱️ Dynamic Timing Control

setTimeout drives typing, deleting, and pause phases precisely.

🎭 Role Rotation

SE Student → Full-Stack Dev → Testing Eng — auto-cycles.

📈 UX Impact

Increases dwell time and demonstrates async DOM manipulation.

UI/UX & Domain-Based Filtering

🎯 Software Engineering Domain Filter

All | JavaScript | HTML/CSS | Testing | SQL

Beyond basic language filters — **Domain Filters** categorize KFUPM academic work (SWE 326, ICS 321) for recruiters.

♿ Accessibility Polish

🔆 High Contrast Dark Mode

Refactored CSS ensures WCAG-compliant contrast ratios in dark theme.

📱 Mobile-First Breakpoints

Responsive layout scales from 320px mobile to 4K widescreen.

Challenges & Problem Solving

01

🌐 Deployment Sync

⚠ **Challenge:** Troubleshooting 404 errors during GitHub Pages migration with broken asset paths.

✓ **Solution:** Standardized all file paths and corrected API endpoint configurations.

02

🎨 UX Accessibility

⚠ **Challenge:** Dark mode text visibility issues on certain background color combinations.

✓ **Solution:** Implemented targeted CSS overrides to enforce readable contrast hierarchy.

03

🧩 Code Cleanliness

⚠ **Challenge:** Managing and orchestrating multiple asynchronous features simultaneously.

✓ **Solution:** Modularized JS functions within a unified DOMContentLoaded listener.

Future Work & Conclusion

Next Steps

Technical Blog

Integrate a Markdown-powered dev blog directly into the portfolio.

SWE Compass Dashboard

Live academic project dashboard with real-time data visualization.

PDF Resume Generator

Automated PDF resume export generated from live web content.

Conclusion

This project successfully bridges the gap between academic theory and professional web delivery.

5+	Tech Layers
3+	Key Features
CI/CD	Deployment